CSM V3 — HARDWARE ORDER SHEET

Wood Base V1.1 Electrical Mounting • Rev V1.0 • 2026-06-04

NAME: ______________________ **ORDER DATE:** ______________________

1. BOLTS (machine screws, partial thread OK, pan or socket head)

☐	QTY	SIZE	USE	NOTES
☐	4	**M5 × 35 mm**	NEMA 23 motor flange	*Use w/ lock washer + threadlock*
☐	4	**M4 × 30 mm**	PSU chassis	*Pan-head; rubber washer optional*
☐	4	**M3 × 30 mm**	Arduino Mega (with standoffs)	*10 mm standoff under PCB*
☐	2	**M3 × 25 mm**	TB6600 stepper driver flange	*Through 4 mm flange tabs*
☐	6	**M3 × 22 mm**	Bucks (×4) + Terminal (×2)	*Or VHB pad on Bucks → skip 4 of these*

Subtotal: 20 bolts (or 16 if using VHB on Bucks)

2. WASHERS

☐	8	**M5 flat washer**	Under motor bolt head AND nut	
☐	4	**M5 lock washer**	Under motor nut (anti-vibration)	*MANDATORY — motor vibrates*
☐	8	**M4 flat washer**	PSU top + bottom	
☐	24	**M3 flat washer**	All M3 bolts, both sides	*Pack of 50 is fine*

3. NUTS

☐	4	**M5 hex nut**	Motor underside	
☐	4	**M4 hex nut**	PSU underside	
☐	12	**M3 hex nut**	All M3 bolts	*Nyloc nut OK — extra security*

4. NYLON STANDOFFS (M3 female-female, threaded)

☐	4	**M3 × 10 mm nylon standoff**	Under Arduino Mega	*PCB airflow*
☐	4	**M3 × 5 mm nylon standoff**	Under each Buck	*Optional if using VHB tape*

5. CONSUMABLES

☐	1	**Loctite 243 (blue)**	Threadlock for M5 motor bolts	*Small bottle is plenty*
☐	1	**3M VHB tape (optional)**	Alt mounting for Bucks	*20×20 mm pads, qty 8*
☐	1	**Pack ferrules 0.5-2.5 mm²**	Wire ends into screw terms	*Crimper required*
☐	1	**Pack heat shrink 3-8 mm**	Wire terminations	*Black + red mixed*

6. DRILL BITS (verify you have these before starting)

☐	1	**Ø 2.0 mm HSS**	Pilot drill for ALL positions	*Prevents wandering*
☐	1	**Ø 3.2 mm HSS**	M3 clearance (10 holes)	*Mega, Bucks, Terminal*
☐	1	**Ø 3.5 mm HSS**	M3 clearance for TB6600 (2 holes)	*TB6600 flange*
☐	1	**Ø 4.5 mm HSS**	M4 clearance for PSU (4 holes)	*PSU chassis*
☐	1	**Ø 5.5 mm HSS**	M5 clearance for motor (4 holes)	*NEMA 23 flange*
☐	1	**Ø 100 mm hole saw OR jigsaw**	Take-down (1 hole)	*Center on datum*
☐	1	**Ø 8 mm HSS (optional)**	Counterbore PSU + motor nuts	*Recess nut into wood*

TIP: A 100-piece M3 + M4 bolt/nut/washer assortment kit usually covers everything except M5 motor bolts.